

NEXORA — Operations Portal

Full-Stack Mini ERP + CRM Operations Suite Documentation

1. Live Deployment & Repository Links

- **Live Frontend URL:** <https://mini-erp-crm-gray.vercel.app/>
- **Live Backend API URL:** <https://mini-erp-crm-puyn.onrender.com/>
- **GitHub Repository:** <https://github.com/Sachinshekhar82/Mini-ERP-CRM.git>
- **Health Check Endpoint:** <https://mini-erp-crm-puyn.onrender.com/api/health>
- **Postman Collection:** [postman/Mini-ERP-CRM.postman_collection.json](#)

2. Demo Test Credentials (All Roles)

- **[ADMIN]** admin@company.com | Password: **password123** (Full system access & User Administration)
- **[SALES]** sales@company.com | Password: **password123** (Customer CRM & Sales Challan creation)
- **[WAREHOUSE]** warehouse@company.com | Password: **password123** (Product catalog & Stock IN/OUT operations)
- **[ACCOUNTS]** accounts@company.com | Password: **password123** (Financial audit & Challan confirmation)

3. System Architecture

NEXORA is built on a decoupled RESTful architecture designed for enterprise B2B wholesale and distribution operations:

- Frontend: React 18 + TypeScript + Vite SPA, styled with custom B2B design tokens, localized skeleton loaders, and frame-0 instant UI shell rendering.
- Backend: Express.js + TypeScript REST API enforcing JWT authentication, Helmet security headers, Zod schema validation, and role-based access control (RBAC).
- Database: Cloud PostgreSQL database (Neon.tech) managed via Prisma ORM with relational integrity and multi-item atomic stock deduction transactions.
- Storage: Sales Challan PDF invoice generation via PDFKit.

4. Core Business Logic & Features

- Atomic Stock Deduction: Multi-item sales challan confirmation checks all product quantities inside a single `prisma.$transaction()`. If any item exceeds available stock, the entire transaction rolls back cleanly with 0 stock changes and 0 movement logs.
- Product Price Snapshot: `SalesChallanItem` retains `unitPriceSnapshot`, `skuSnapshot`, and `productNameSnapshot` at creation time, ensuring future catalog price edits never corrupt historical invoices.
- Customer CRM Pipeline: Multi-field search (name, business, email, phone, GSTIN), filters, pagination, and chronological follow-up timeline notes.
- Inventory & Low Stock Warnings: Threshold alerts (`currentStock <= minimumStock`) and movement audit logs.

5. Known Limitations & Future Scope

- Known Limitation: Sales challan cancellation currently applies to DRAFT status. Cancelling a CONFIRMED challan requires issuing a manual reverse Stock IN adjustment to preserve audit integrity.
- Future Improvement: Multi-currency support and automated email dispatch of PDF invoices directly to customer email addresses.